

2. Storage Tank Bottom Plate Thickness Assessment

To: Mechanical Integrity Department, Operations **From:** L. Fennimore, Mechanical Integrity Engineer
Date: 2026-06-15 **Subject:** Tank TK-118 — API 653 Bottom Plate MRT Assessment Result (per SOP-TK-140 Clause 5.1)

Background

Per SOP-TK-140 Section 5, Clause 5.1, floor plate minimum remaining thickness (MRT) for atmospheric storage tanks must be evaluated against API 653 criteria following any internal floor-scan inspection. The robotic MFL floor scan completed this cycle on TK-118 triggered the Clause 5.4 engineering-review threshold due to isolated pitting exceeding the general corrosion trend.

Assessment Result

The calculated minimum required bottom plate thickness for TK-118, based on product specific gravity, tank diameter, and API 653 Annex B criteria, is **2.3 mm**. This value supersedes the placeholder MRT figure carried in the prior inspection interval planning sheet.

Any floor-scan reading at TK-118 below 2.3 mm shall be treated as a localized fitness-for-service exceedance: the affected plate area shall be flagged for repair or patch-plating prior to tank return to service, consistent with SOP-TK-140 Clause 5.3's requirement that no out-of-service tank be returned to service until all sub-MRT indications are dispositioned.

Distribution

Filed to the Tank Integrity File per SOP-TK-140 Clause 7.1. Retain for the operating life of the tank plus 10 years.